

Internet of Things

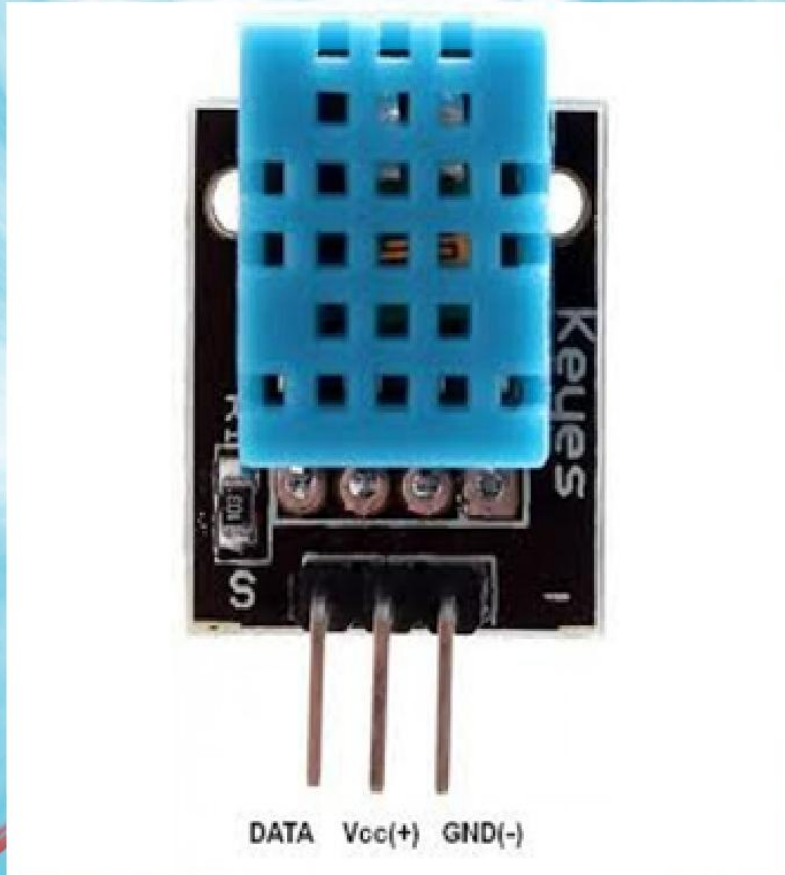
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Problem

I want to find out the Room Temperature what sensor should I choose?





Its DHT



DHT Sensor

DHT stands for **Digital Humidity and Temperature**. The DHT sensor is a low-cost digital sensor for sensing temperature and humidity.

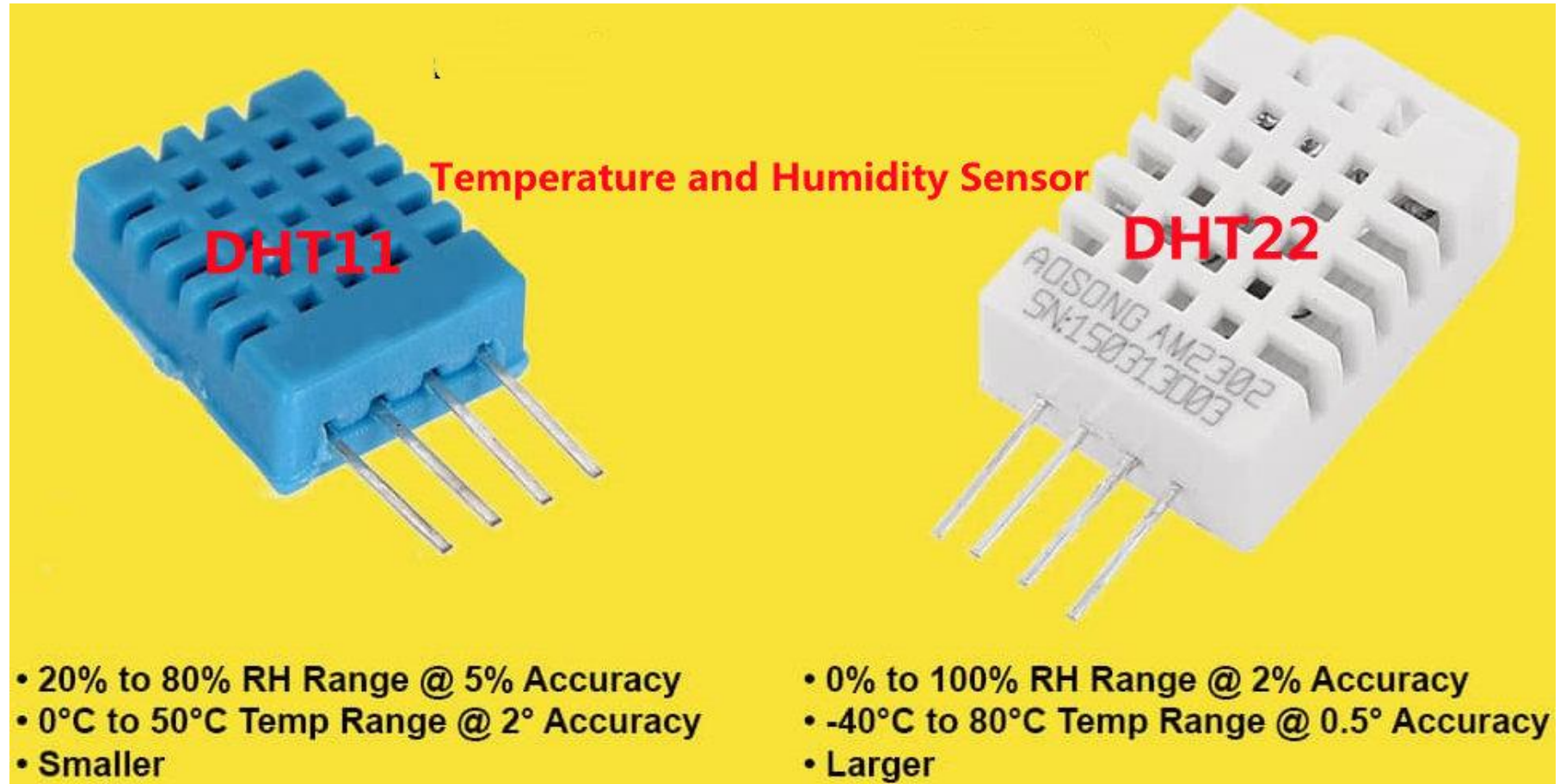
This sensor can be easily interfaced with any micro-controller such as Arduino, Raspberry Pi to measure humidity and temperature instantaneously.

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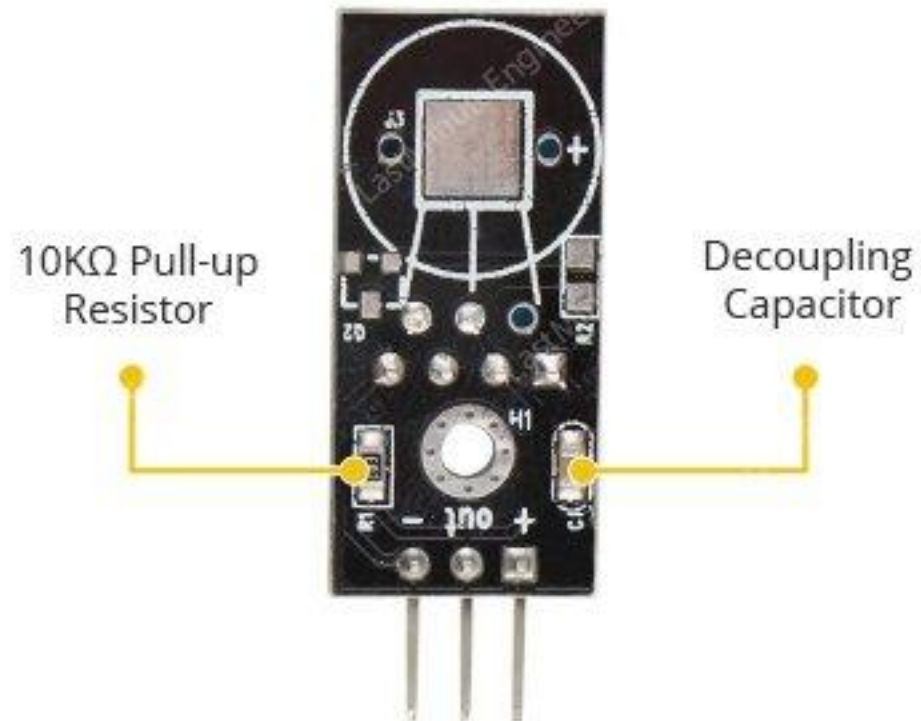
We have two versions of the DHT sensors they are:

1. DHT11
2. DHT22

DHT 11 vs DHT 22



Supporting Circuitry



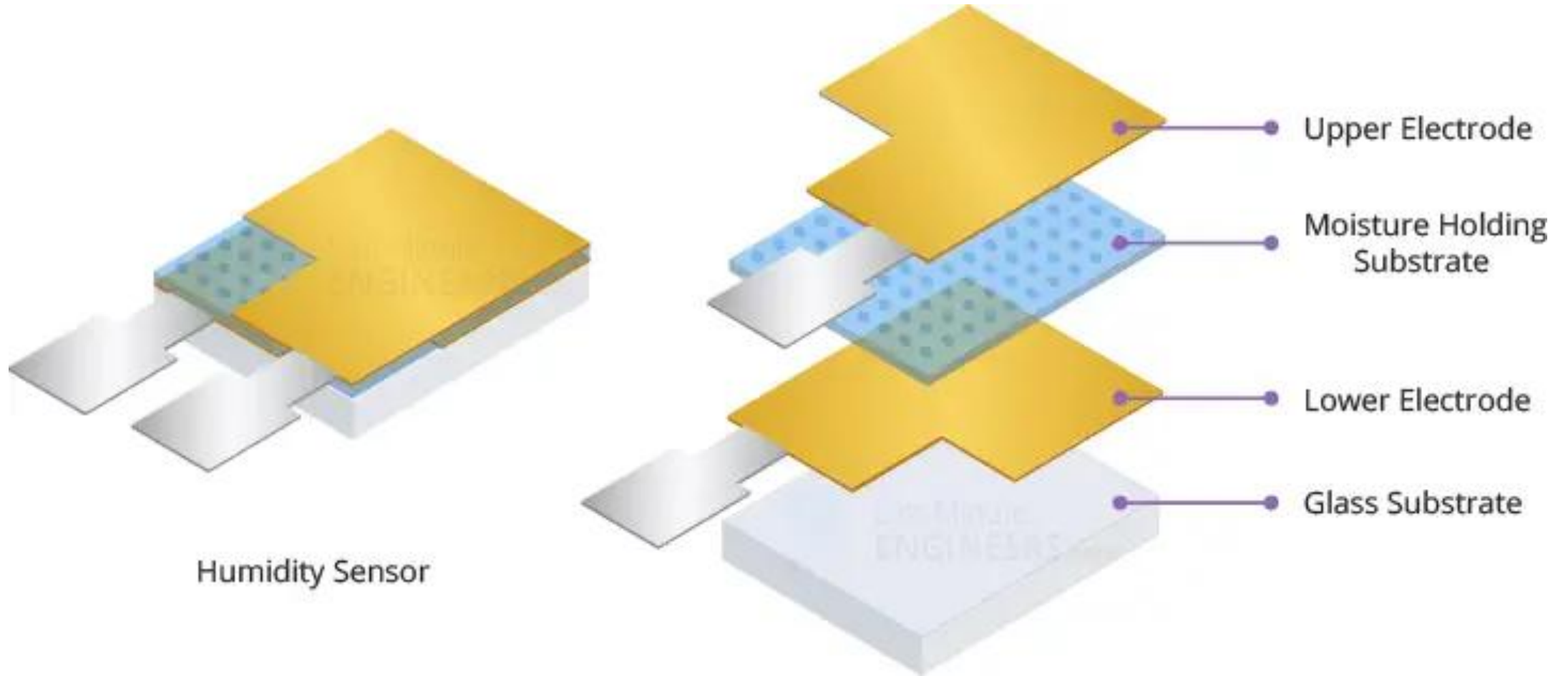
Inside the Sensor



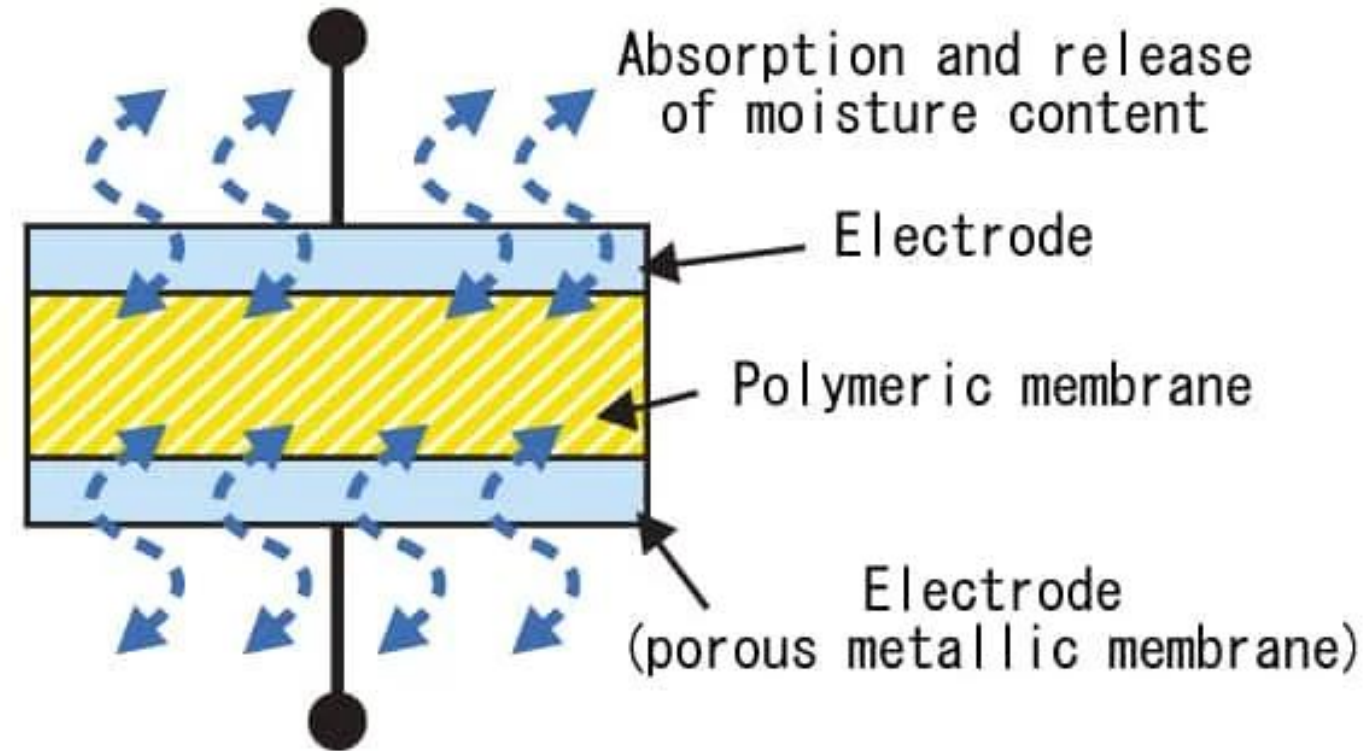
Contd.

- If you remove the sensor's casing, you will find an NTC thermistor and a humidity sensing component inside.
- The humidity sensing component has two electrodes with a moisture-holding substrate (usually a salt or conductive plastic polymer) in between.
- As the humidity rises, the substrate absorbs water vapor, resulting in the release of ions and a decrease in the resistance between the two electrodes.
- This change in resistance is proportional to the humidity, which can be measured to estimate relative humidity.

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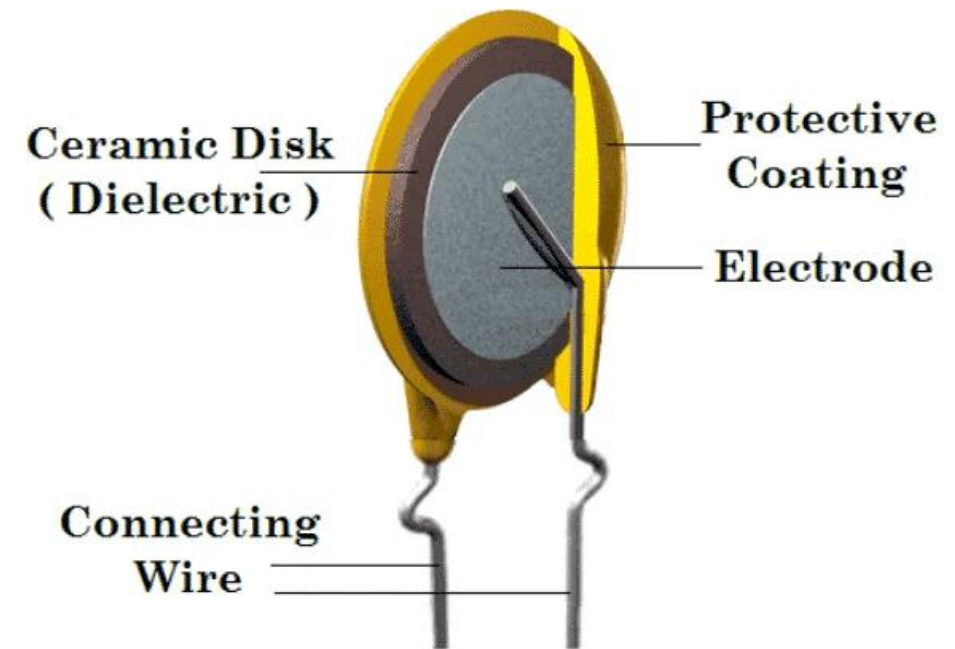


Humidity and Temperature Sensor



Principle of polymeric
membrane humidity sensor

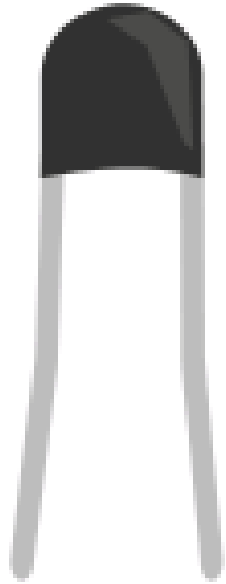
Humidity Sensor



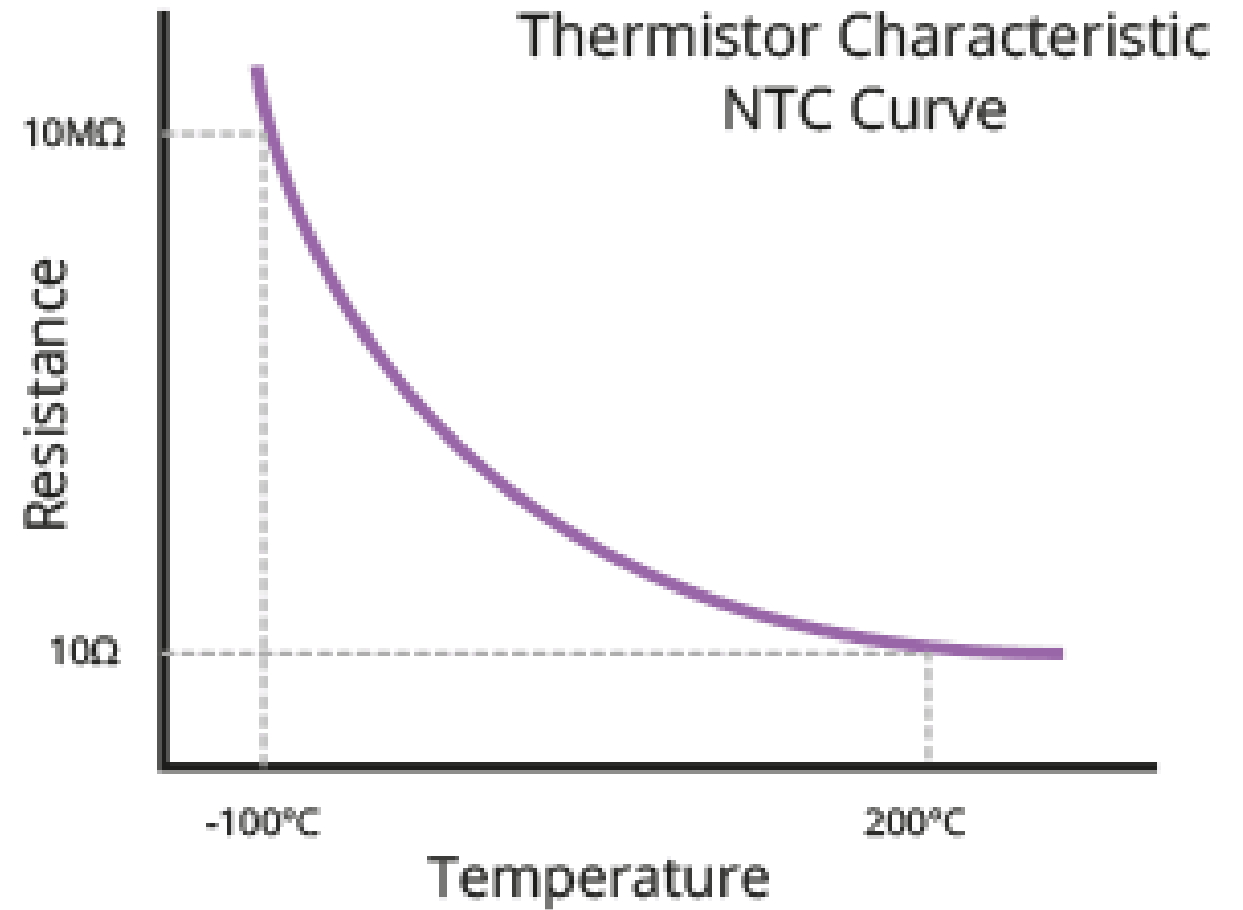
Temperature Sensor

Contd.

- DHT11 also includes a NTC thermistor for measuring temperature. A thermistor is a type of resistor whose resistance varies with temperature.
- Technically, all resistors are thermistors in the sense that their resistance changes slightly with temperature, but this change is typically very small and difficult to measure. Thermistors are designed so that their resistance changes dramatically with temperature (by 100 ohms or more per degree). The term “NTC” stands for “Negative Temperature Coefficient,” which means that resistance decreases as temperature rises.



NTC Thermistor



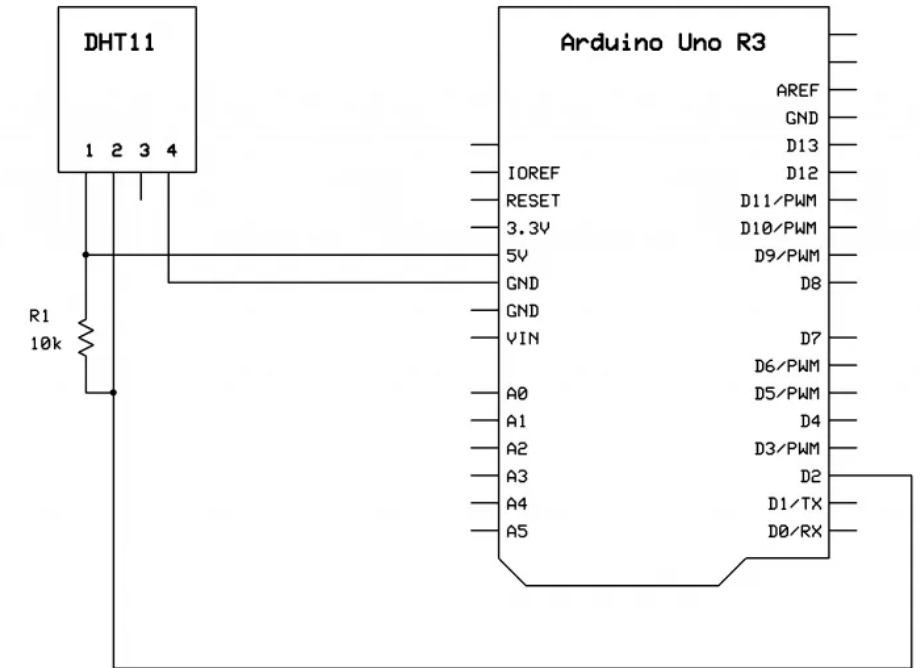
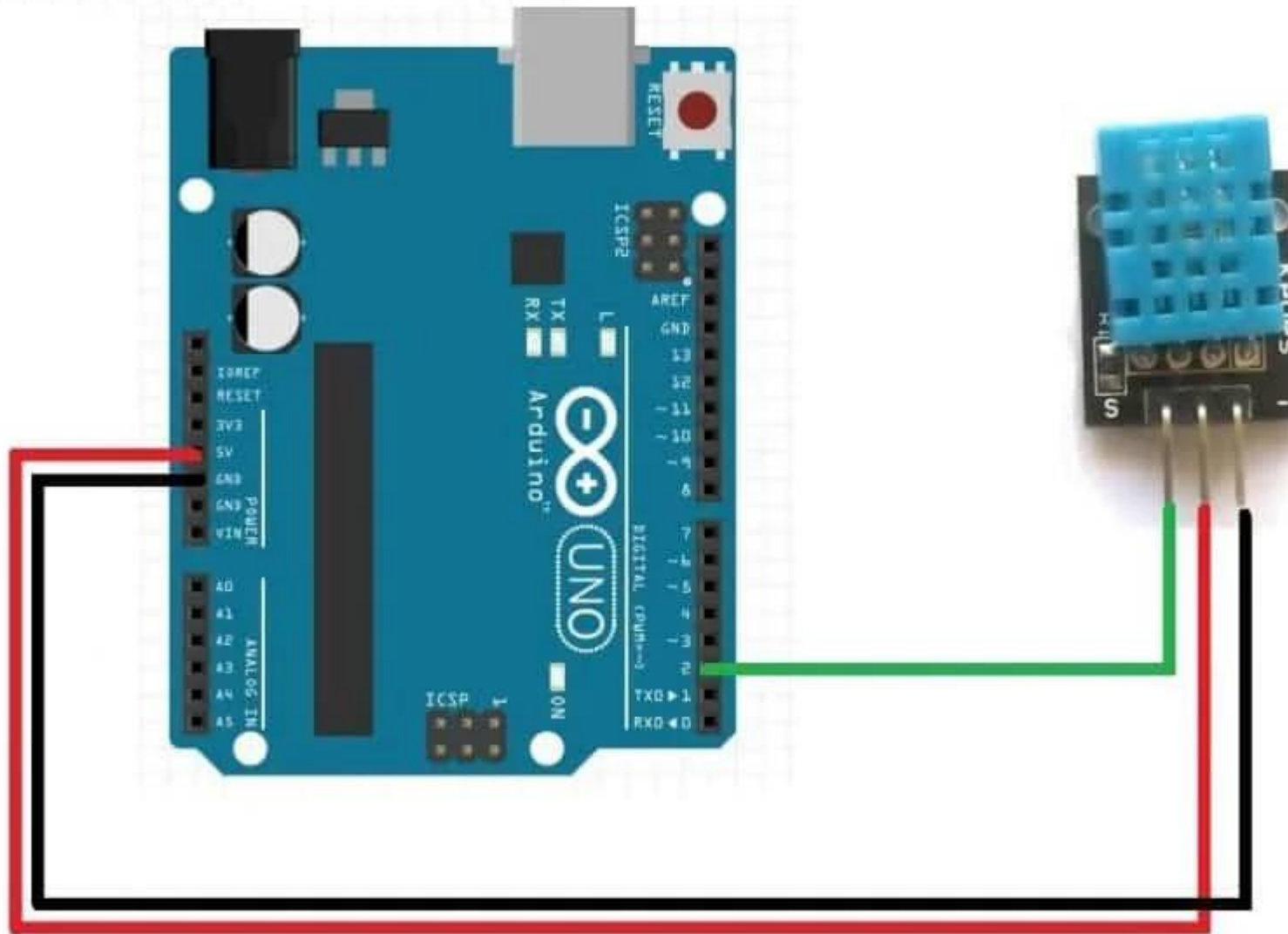
Corelation between Resistance and Temperature

Contd.

$$B_{(T_1/T_2)} = \frac{T_2 \times T_1}{T_2 - T_1} \times \ln \left(\frac{R_1}{R_2} \right)$$

- $B_{(T_1/T_2)}$ (**B-constant / Beta Value**): A material constant given in **Kelvin (K)** that describes the sensitivity of the thermistor material over a specific temperature range between T_1 and T_2 . Typical values range between 3000 K and 5000 K.
- T_1 : The baseline reference temperature in **Kelvin (K)** (usually $25^\circ\text{C} = 298.15\text{ K}$).
- T_2 : The second target temperature in **Kelvin (K)**.
- R_1 : The electrical resistance of the thermistor at temperature T_1 (measured in Ohms, Ω).
- R_2 : The electrical resistance of the thermistor at temperature T_2 (measured in Ohms, Ω).
- $\ln$: The natural logarithm function.

Connections



Installing DHT Library

- Go to Manage Library.
- **Sketch-> Include Library-> Manage Libraries**
- Install **dhtlib**.

Bonus Code (may or may not work)

```
1. void setup()
2. {
3.   Serial.begin(9600);
4. }
5. void loop()
6. {
7.   Serial.println( readTemp(), DEC );
8.   delay(1000);
9. }
10. long readTemp()
11. {
12.   long result; // Read temperature sensor against 1.1V reference
13.   ADMUX = _BV(REFS1) | _BV(REFS0) | _BV(MUX3); delay(2); // Wait for Vref to settle
14.   ADCSRA |= _BV(ADSC); // Convert
15.   while (bit_is_set(ADCSRA,ADSC));
16.   result = ADCL;
17.   result |= ADCH<<8;
18.   result = (result - 125) * 1075; return result;
19. }
```

Assignment

- 1. To interface DHT to find the Humidity.**
- 2. To interface DHT to find the Temperature.**
- 3. Convert the Temperature from Celsius to Fahrenheit.**

Please write the same in your Lab Record Books with the Circuit Diagram on the blank page, and upload your codes to Teams as .ino for each assignment. Please sign each code with your name and roll number in the comments.

Thanks